



Multistage Universal Speech Enhancement System for URGENT Challenge

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Abstract

During audio transmission and processing, various distortions may occur. To effectively address this challenge, we developed a multistage universal speech enhancement system, consisting of four submodules, namely audio declipping, packet loss compensation, audio separation, and spectral inpainting. These modules operate across the time, sub-band, and time-frequency domains. We employed a pretrain-finetune training paradigm and introduce a self-distillation method to further improve performance. Experiments on large-scale datasets demonstrate that our system outperforms in multiple evaluation metrics, particularly in improving subjective speech quality. The proposed system ranked 1st in the URGENT 2024 challenge with a MOS of 3.52 and placed 4th in the second track of the URGENT 2025 challenge.

Index Terms: Universal Speech Enhancement, Multistage, URGENT Challenge

1. Introduction

In audio encoding, transmission, and processing, various distortions such as noise, signal amplitude clipping, codec artifacts, and spectral loss are commonly encountered. These distortions significantly impair speech perceptual quality and lead to performance degradation in downstream tasks, such as automatic speech recognition. Recent speech enhancement systems designed for a single task have achieved remarkable success, producing significant results in specific scenarios, including speech enhancement [1, 2], spectral band extension [3, 4], and packet loss compensation (PLC) [5]. Some models employ multistage [6, 7] or end-to-end [8] approaches to simultaneously address multiple of the aforementioned distortions. Universal speech enhancement [9], which aims to handle all potential disturbances, has gained increasing attention in recent research.

To extend speech enhancement capabilities from single to universal scenarios, it is essential to address significant variations in distortions across multiple scales and domains. Early works attempted to integrate speech enhancement with dereverberation [10], as these tasks can be equivalently formulated as additive noise removal through spectral mask estimation. Unlike additive noise, spectral band loss is more challenging, as it requires spectrogram estimation, which is difficult to learn. Unified approaches such as TF-GridNet [2] or Apollo [3] have been proposed to simultaneously handle noise and spectral distortions. Discontinuities in the time domain are another critical factor that contributes to perceptual degradation, commonly manifested as audio clipping, interruptions, and packet loss. Conventional time-frequency models struggle to accurately localize distortions at signal sampling points due to their limited spectral resolution, making time domain [11] and subband do-

main [5] models more widely used. Multistage models [7] provide a robust solution by combining the advantages of different models or domains to effectively handle various distortions. These models have been extensively studied in the context of speech enhancement and restoration [6, 12], demonstrating their compatibility with both noise reduction and spectrogram extension. Generative models, such as diffusion models [13, 14] and generative adversarial networks (GANs) [15], have also been introduced to the field of universal speech enhancement. These approaches typically treat the correction of various distortions as a unified generative task, conditioned on the distorted speech feature. In recent studies, how semantic information embedded in speech can facilitate audio restoration has been investigated [16, 17]. Using pre-trained self-supervised embeddings, audio restoration models can produce more intelligible speech, even under severe distortion or packet loss conditions [18].

Although previous competitions such as the Deep Noise Suppression (DNS) Challenge [19] have substantially increased attention and research on the speech enhancement task, there is a dearth of relevant datasets and appropriate evaluation metrics. Researchers often focus on closed-domain datasets from specific subfields that lack comprehensive cross-domain validation. Furthermore, there has been no well-recognized benchmark to assess the performance of these models across the wide variety of speech quality distortions encountered in real-world scenarios. The URGENT Challenge¹ provides such a platform, encouraging researchers to develop universally robust speech enhancement systems. More importantly, the organizers offer 14 evaluation metrics in four main categories, evaluating objective signal quality, subjective human perception, and downstream task performance.

This paper presents our universal speech enhancement system submitted to the URGENT Challenge. The system addresses various types of distortion through a combination of four submodules, which respectively handle clipping, packet loss, additive noises, and spectrogram distortion. To optimize each submodule for optimal performance, we adopted a training paradigm in which the submodules are individually pretrained, followed by fine-tuning of the entire system. During testing, we also introduced a routing strategy by detecting the existence of distortions, where an additional detector controls the activation of the Declipping and PLC modules. Our system achieved the **1st** place in the URGENT Challenge 2024, with a mean opinion score (MOS) of 3.52, significantly higher than other teams. Additionally, we ranked **4th** in Track-2 of URGENT Challenge 2025. These results demonstrate the outstanding performance and effectiveness of our system in addressing complex audio distortions.

¹<https://urgent-challenge.com/>

2. System

2.1. Problem formulations and Methodology

Let $\mathbf{s} \in \mathbb{R}^T$ represent the clean speech signal and the degraded speech $\mathbf{x}$ can be written as

$$\mathbf{x} = d_1 \circ d_2 \circ \dots \circ d_N(\mathbf{s}) \in \mathbb{R}^T \quad (1)$$

where $\circ$ stands for the composition of the function and N is the number of distortions. The distortions in real scenarios can be mainly classified into the following categories.

Clipping: $d_{clip}(\mathbf{s}) = \max(\min(\mathbf{s}, \delta_1), \delta_2)$, where δ_1 and δ_2 are the lower and upper bounds of the signal that limit the amplitude. In this paper, we only consider hard clipping and do not consider the problem of sample discontinuity.

Packet loss: $d_{PL}(\mathbf{s}) = \mathbf{s} \odot \mathbf{m}$, where $\mathbf{m} \in \mathbb{R}^T$ denotes the mask of packet loss in the time domain. The value of $\mathbf{m}(k)$ is 0 or 1, which indicates whether packet loss occurred in packet k .

Additive noise and reverberation: $d_{noise}(\mathbf{s}) = \mathbf{s} * \mathbf{h} + \mathbf{n}$, where $\mathbf{h}$ and $\mathbf{n}$ respectively denote the room impulse response (RIR) and the noise signal. In this paper, clean speech with early reverberation is used as the noise suppression target.

Low bandwidth: $d_{lowb}(\mathbf{s}) = \text{Resample}(\mathbf{s} * \mathbf{h}_{lp}, sr_o, sr_t)$, where $\mathbf{h}_{lp}$ is a low-pass filter and $\text{Resample}(\cdot, sr_o, sr_t)$ refers to the resampling operation that converts the original sample rate sr_o to the target sr_t .

Spectral artifacts: $d_{spec}(\mathbf{s}) = \mathbf{F}(\mathbf{s})$, where $\mathbf{F}(\cdot)$ denotes the process of a specific audio codec or front-end algorithm.

During the training process, we do not assume a fixed distortion order for data construction. Instead, during enhancement, we prioritize handling time-domain distortions, such as clipping, followed by frequency-domain distortions. This approach facilitates the decoupling of the effects from different domains, thereby reducing the optimization difficulties. Spectrum inpainting is placed at the final stage to mitigate the additional artifacts introduced by the preceding stages. Based on this methodology, we construct a multistage audio enhancement system consisting of submodules, denoted E_i . The enhanced speech $\mathbf{e} \in \mathbb{R}^T$ is represented as

$$\mathbf{e} = E_1 \circ E_2 \circ \dots \circ E_M(\mathbf{x}) \quad (2)$$

2.2. System overview

As shown in Figure 1, our system addresses distortions by utilizing a combination of four submodules: a time-domain declipping module, a time-frequency domain speech separation module, a subband-domain PLC model, and a time-frequency domain inpainting module. These modules are cascaded and can be independently optimized. All submodules follow an encoder-decoder architecture. The structure of the declipping module is depicted in Figure 2-(a). We adopted the Demucs architecture [20], which consists of 12 layers of symmetric convolutional encoders and decoders, along with a bottleneck layer comprising 3 BLSTMs. The PLC module is based on BS-PLCNET [5], the champion system of the PLC Challenge 2024 [21]. This model operates in the pseudo quadrature mirror filter bank (PQMF) domain and employs multi-resolution adversarial training for learning. Given that additive noise is the most common type of distortion, we employ the state-of-the-art audio separation framework BSRoformer [22] for denoising and dereverberation. As shown in Figure 2-(b), this model separates speech by dividing the spectrum into 62 non-overlapping frequency bands, which are processed through 16

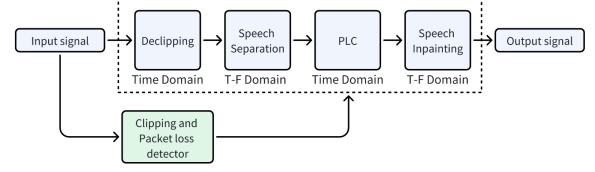


Figure 1: Schematic diagram of the proposed system.

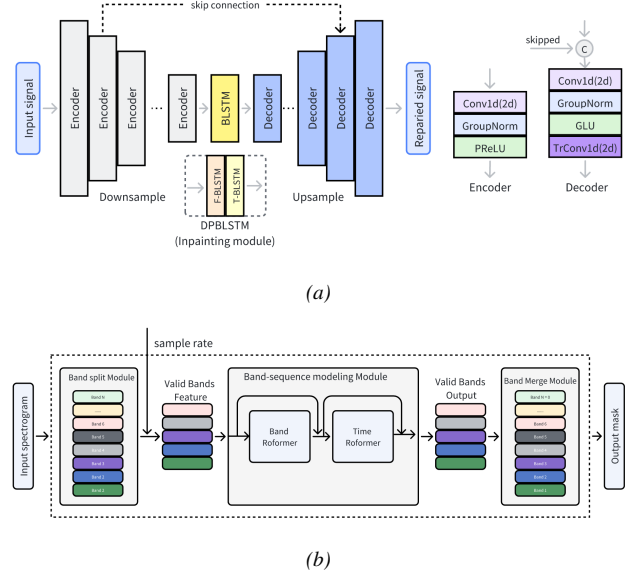


Figure 2: Structure details of the declipping and inpainting module (a) and the separation module (b).

Rope-Transformer layers [23] to capture the dependencies between the bands and the frames. The speech spectral inpainting module, similar to the declipping model, employs a 2-dimensional convolutional recurrent network in the time-frequency domain, in which the bottleneck BLSTM is replaced by DPBLSTM [24]. To process audio with arbitrary sample rates, both the declipping and the separation module are trained on a diverse range of sample rates, spanning from 8 kHz to 48 kHz. Specifically, the sample rate is provided as an additional input to the BSRoformer following the band-split module, enabling it to determine the appropriate number of frequency bands for subsequent processing. For the PLC and inpainting modules, all the input clips are resampled to 48 kHz and the outputs are then resampled back to the original sample rate.

To further reduce the inference complexity, we trained a time-domain detector which can assess the occurrence of clipping and packet loss, and then decide whether to activate the declipping and PLC module accordingly. By using Temporal Convolutional Networks (TCN), the detector effectively identifies clipping and packet loss at the sample level. The model is composed of 7 convolutional layers followed by a linear layer for probability prediction. The predictions generated by the model not only determine whether the module should be activated but also act as supplementary input features to boost model performance.

Table 1: Evaluation results on the non-blind test set of URGENT Challenge 2024

	Non-intrusive metrics		Intrusive metrics				Downstream task metrics	
Metrics	DNSMOS	NISQA	PESQ	ESTOI	SDR	LSD	SpeechBERT Score	WAcc
Base (Declicking + Separation)	3.21	3.86	2.71	0.89	16.94	6.30	0.87	82.81
+ Inpainting	3.22	4.01	2.88	0.90	16.79	2.80	0.88	82.24
+ Self-distillation (2nd)	3.26	4.15	2.92	0.90	16.84	2.84	0.88	82.19
Team A (1st)	3.20	4.16	3.04	0.90	16.65	2.70	0.89	83.31
Team B (3rd)	3.21	4.14	2.88	0.89	16.05	2.97	0.88	83.00

Table 2: Evaluation results on the blind test set of URGENT Challenge 2024

	Non-intrusive metrics		Intrusive metrics				Downstream task metrics		Subjective metrics
Metrics	DNSMOS	NISQA	PESQ	ESTOI	SDR	LSD	SpeechBERT Score	WAcc	MOS
Our system (1st)	3.06	3.66	2.65	0.87	14.58	2.92	0.84	73.57	3.52
Team A (2nd)	3.00	3.59	2.80	0.87	14.52	2.78	0.85	72.91	3.46
Team B (4th)	2.95	3.35	2.66	0.86	13.54	2.70	0.85	73.10	3.40

2.3. Training framework

We followed the data construction approach from the URGENT Challenge, expressed as equation (1). We also generated parallel augmented samples for each stage of the model. To optimize each module to its best, we adopted a training paradigm where the submodules are pre-trained individually and then fine-tuned as a whole system. During the pre-training phase, we categorized the modules as discriminative or generative models and applied different training losses for each group. For the de-clicking and separation models, we used a discriminative loss $\mathcal{L}$, which is the weighted sum of the time-domain loss $\mathcal{L}_T$ and frequency-domain loss $\mathcal{L}_F$ with a weight $\lambda = 0.5$, expressed as

$$\mathcal{L} = \lambda \times \mathcal{L}_T(s, \hat{s}) + (1 - \lambda) \times \mathcal{L}_F(\mathbf{S}, \hat{\mathbf{S}}) \quad (3)$$

where the time-domain loss is based on L1 loss and the frequency-domain loss uses multi-resolution compressed spectral MSE loss [25] as

$$\mathcal{L}_T(s, \hat{s}) = \frac{1}{N} \sum_{t=1}^N |s(t) - \hat{s}(t)| \quad (4)$$

$$\mathcal{L}_F(\mathbf{S}, \hat{\mathbf{S}}) = \sum_{nfft \in \mathcal{S}} \text{MSE}_{nfft} \left(\frac{\mathbf{S}}{|\mathbf{S}|^{0.5}}, \frac{\hat{\mathbf{S}}}{|\hat{\mathbf{S}}|^{0.5}} \right) \quad (5)$$

where $\mathbf{S}$ and $\hat{\mathbf{S}}$ represent the clean and estimated spectrograms, respectively, using an FFT with a specified length $nfft$. In our experiments, the selected set of FFT length was configured as $\mathcal{S} = \{256, 512, 1024, 2048, 4096\}$. We employ adversarial generative training for the PLC and the spectral inpainting module, introducing a multi-resolution frequency-domain discriminator D_f and a multi-resolution subband-domain discriminator D_s . The discriminators aim to distinguish between the predicted $\hat{s}$ and the real target s , by minimizing

$$\mathcal{L}_d = \mathbb{E}_{s \sim p_{data}} [(D_f(s) - 1)^2 + (D_s(s) - 1)^2 + (D_f(\hat{s}))^2 + (D_s(\hat{s}))^2] \quad (6)$$

The inpainting model, serving as a generator, utilizes the weighted sum of the spectral loss $\mathcal{L}_{spec}$ and the GAN loss $\mathcal{L}_{gan}$,

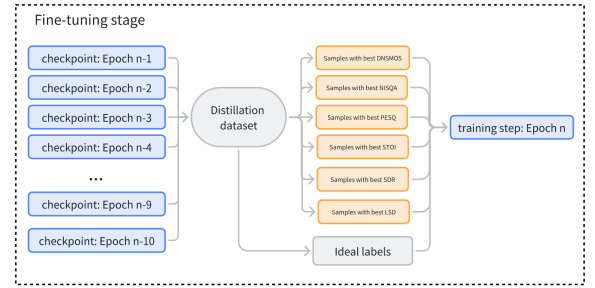


Figure 3: Schematic of the self-distillation method used in the fine-tuning phase.

expressed as

$$\mathcal{L}_g = \mathcal{L}_{gan} + \alpha \times \mathcal{L}_{spec} \quad (7)$$

$$\mathcal{L}_{spec} = \mathcal{L}_F(\mathbf{S}, \hat{\mathbf{S}}) \quad (8)$$

$$\mathcal{L}_{gan} = \mathbb{E}_{s \sim p_{data}} [(D_f(\hat{s}) - 1)^2 + (D_s(\hat{s}) - 1)^2 + \beta \times (\mathcal{L}_{feat}(s, \hat{s}; D_f) + \mathcal{L}_{feat}(s, \hat{s}; D_s))] \quad (9)$$

where $\mathcal{L}_{feat}$ is the MSE loss of the outputs from the hidden layers of the discriminators, which contributes to more stable training. The weights α and β are set to 45 and 2, respectively.

During the fine-tuning stage of the entire system, we introduce a metric-based self-distillation approach to enhance model performance across multiple evaluation metrics, shown as Figure 3. Specifically, at each epoch, we evaluate the output of the model on a constructed distillation dataset, which is a subset of the original training data. For each metric, we identify the best-performing samples from the last 10 checkpoints. These high-quality enhanced samples serve as additional supervision labels for subsequent training, allowing the model to integrate the strengths of different checkpoints and optimize its performance across various metrics in a more balanced manner. In our experiments, the selected samples are mixed with ideal labels at a ratio of 30% during fine-tuning.

Table 3: Evaluation result on the URGENT challenge 2025 blind test set (track 2)

	Non-intrusive metrics		Intrusive metrics				Downstream task metrics		Subjective metrics
Metrics	DNSMOS	NISQA	PESQ	ESTOI	SDR	LSD	SpeechBERT Score	WAcc	MOS
Team B (1st)	2.92	3.24	2.47	0.79	11.10	2.99	0.84	76.06	3.32
Team C (2nd)	2.90	3.11	2.44	0.79	11.61	3.44	0.83	77.51	3.27
Team D (3rd)	2.83	2.90	2.49	0.81	11.76	3.00	0.85	78.40	2.99
Our system (4th)	2.91	3.31	2.31	0.78	11.21	3.57	0.83	76.77	3.25
+ Detector	2.91	3.28	2.88	0.78	11.10	3.78	0.83	77.15	-

3. Results

3.1. Parameter setups and Metrics

In our experiments, we utilized the full dataset provided by the URGENT Challenge, generating nearly 2 TB of FLAC-format audio for training. The four modules have parameter sizes of 6 M, 125 M, 3.8 M and 4.6 M, with a total computational cost of 350 GMACs for 48 kHz input. The model was trained on 8 A100-80GB GPUs, totaling 44 GPU days. The learning rate was set to $5e-4$ for pre-training and $1e-5$ for fine-tuning, with a decay rate of 0.95, both optimized with Adam. Self-distillation was applied for 20 epochs. For evaluation, we selected 8 key metrics from the original challenge-provided evaluation metrics, covering non-intrusive speech quality metrics (DNSMOS [26] and NISQA [27]), intrusive metrics (PESQ [28], ESTOI [29], SDR [30] and LSD [31]), downstream task performance (SpeechBERT Score [32] and WAcc) and subjective evaluation (MOS).

3.2. Experimental Results

We developed multiple systems and evaluated them on the URGENT Challenge 2024 non-blind test set. As shown in Table 1, audio after declipping and separation module already achieved good DNSMOS and SDR scores, but the objective metrics were suboptimal due to spectral loss. Notably, the inpainting module significantly improved these metrics, demonstrating the advantage of multi-stage processing in integrating the strengths of different modules. Furthermore, with the addition of self-distillation, most metrics showed improvement, particularly the non-intrusive ones, while a few, such as SDR and WAcc, exhibited slight degradation. This suggests that the approach effectively balances trade-offs among different metrics. A comparison with other top-tier systems of URGENT Challenge 2024, as presented in Table 1 and 2, highlights the superiority of our system in non-intrusive metrics and perceptual quality. Ultimately, our system ranked first across five metrics, notably achieving a MOS of 3.52, significantly outperforming other teams. This contributed to our top ranking on the blind test set. However, since our approach did not primarily focus on downstream tasks during training and fine-tuning, our performance on those metrics was relatively lower.

The results of our system with an additional PLC module on the URGENT 2025 blind test set are presented in Table 3. Although the system achieved robust performance on non-intrusive metrics, severe packet loss in some samples posed challenges beyond the current capabilities of our modules, resulting in a decline in objective scores. This is likely attributable to the absence of a PLC module specifically designed for the competition task. We also evaluated the activation strategy of the detector. In our experiments, the declipping and PLC modules are selectively disabled when the clipped sample ratio falls

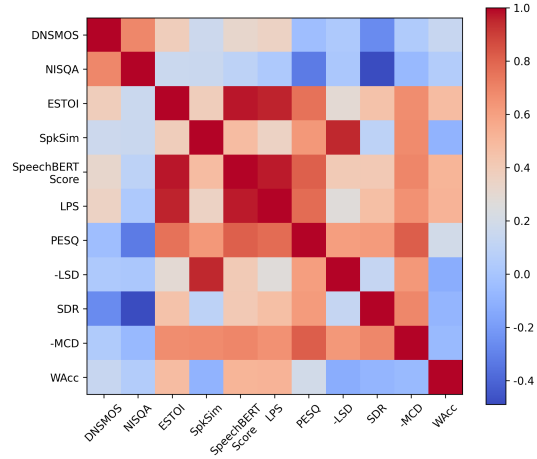


Figure 4: Visualization of the Pearson correlation matrix for all metrics on the URGENT Challenge 2024 non-blind test set.

below 5% or in the absence of packet loss events. The results show that this approach effectively reduces computational cost without significantly affecting performance.

Moreover, we visualized the Pearson correlation matrix for all metrics across our 38 submissions in Figure 4. A positive correlation is observed among the four main metric categories. Notably, non-intrusive metrics exhibit relatively lower correlation with other metrics, making them effective references for speech quality evaluation. Within intrusive metrics, LSD, SDR, and MCD [33] also demonstrate strong independence. For downstream tasks, SpeechBERTScore and ESTOI show strong correlations, while WAcc demonstrates limited correlation with other metrics and can be considered as an independent reference. These insights can serve as valuable guidelines for speech quality evaluation in future work.

4. Conclusion

In summary, we propose a multistage cross-domain model for the universal speech enhancement task, trained within a pretrain-finetune framework. By optimizing both objective and subjective evaluation metrics, our system achieves top-tier performance and ranks 1st in the URGENT Challenge 2024 and 4th in the URGENT Challenge 2025. In the future, we aim to improve generalization and computational efficiency of the model and leveraging more generative modules to further boost performance, especially under conditions of severe distortions.

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